Chapter 14

Wireless LAN

Wireless LAN Problem

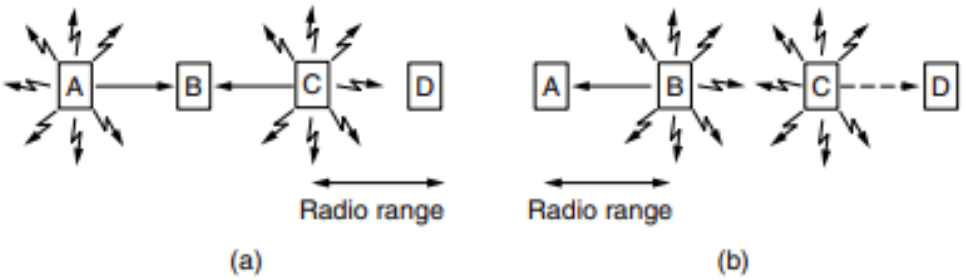


Figure 4-11. A wireless LAN. (a) *A* and *C* are hidden terminals when transmitting to *B*. (b) *B* and *C* are exposed terminals when transmitting to *A* and *D*.

Wireless LAN Problem

1. Hidden Terminal Problem

2. Exposed Terminal Problem

Wireless LAN Protocols

MACA (Multiple Access with Collision Avoidance)
MACAW (Multiple Access with Collision Avoidance for wireless)

RTS: Request to send
CTS: Clear to send

MACA

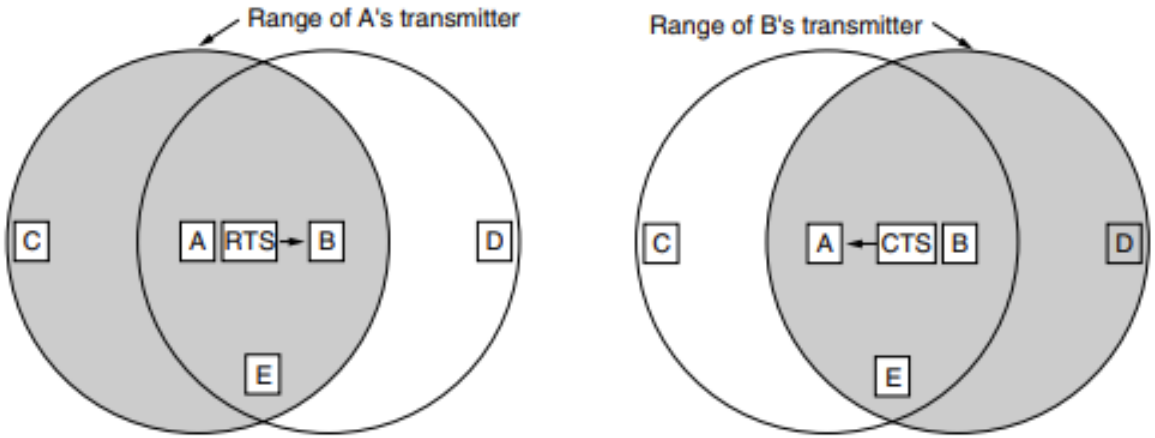


Figure 4-12. The MACA protocol. (a) *A* sending an RTS to *B*. (b) *B* responding with a CTS to *A*.

Basic Service Set (BSSs)

A BSS without an AP is called an adhoc network;

A BSS with an AP is called an infrastructure network.

Figure 14.1 *Basic service sets (BSSs)*

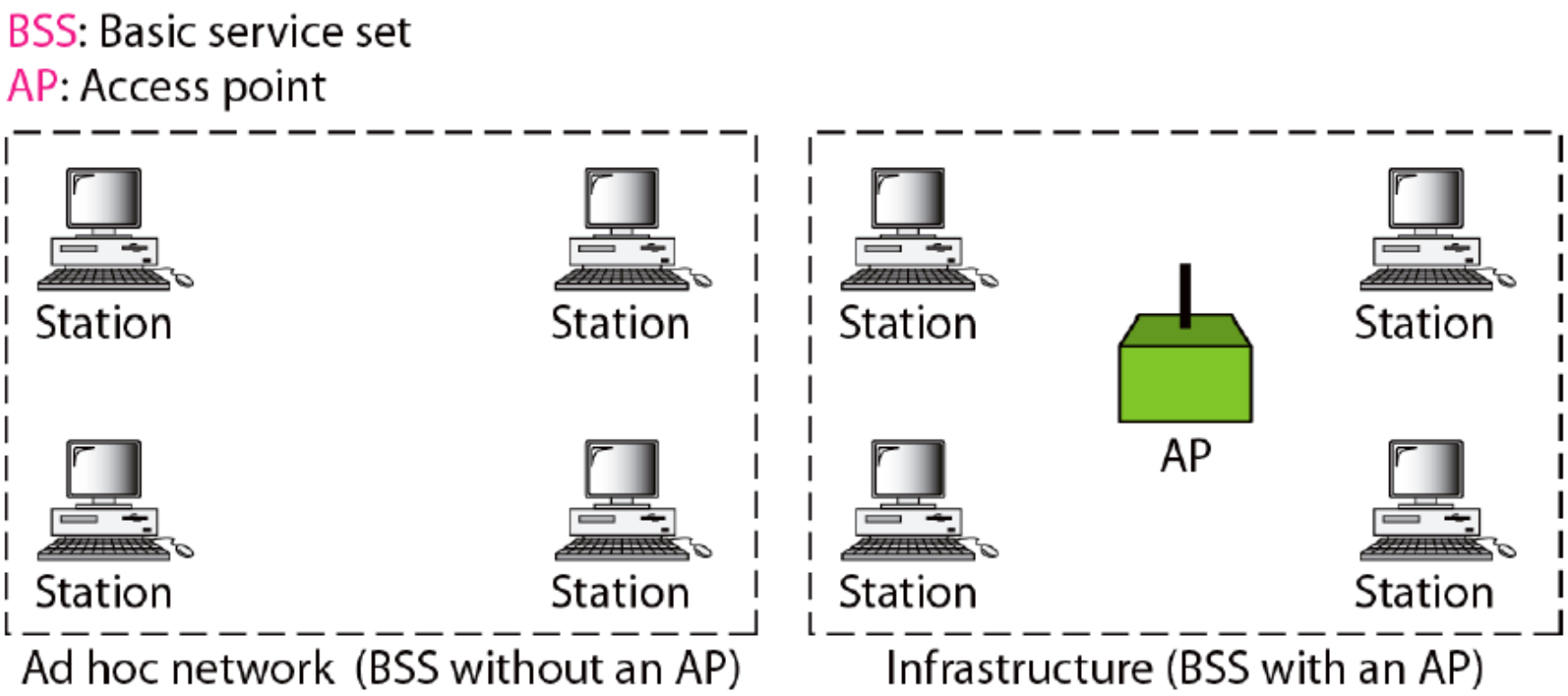


Figure 14.2 *Extended service sets (ESSs)*

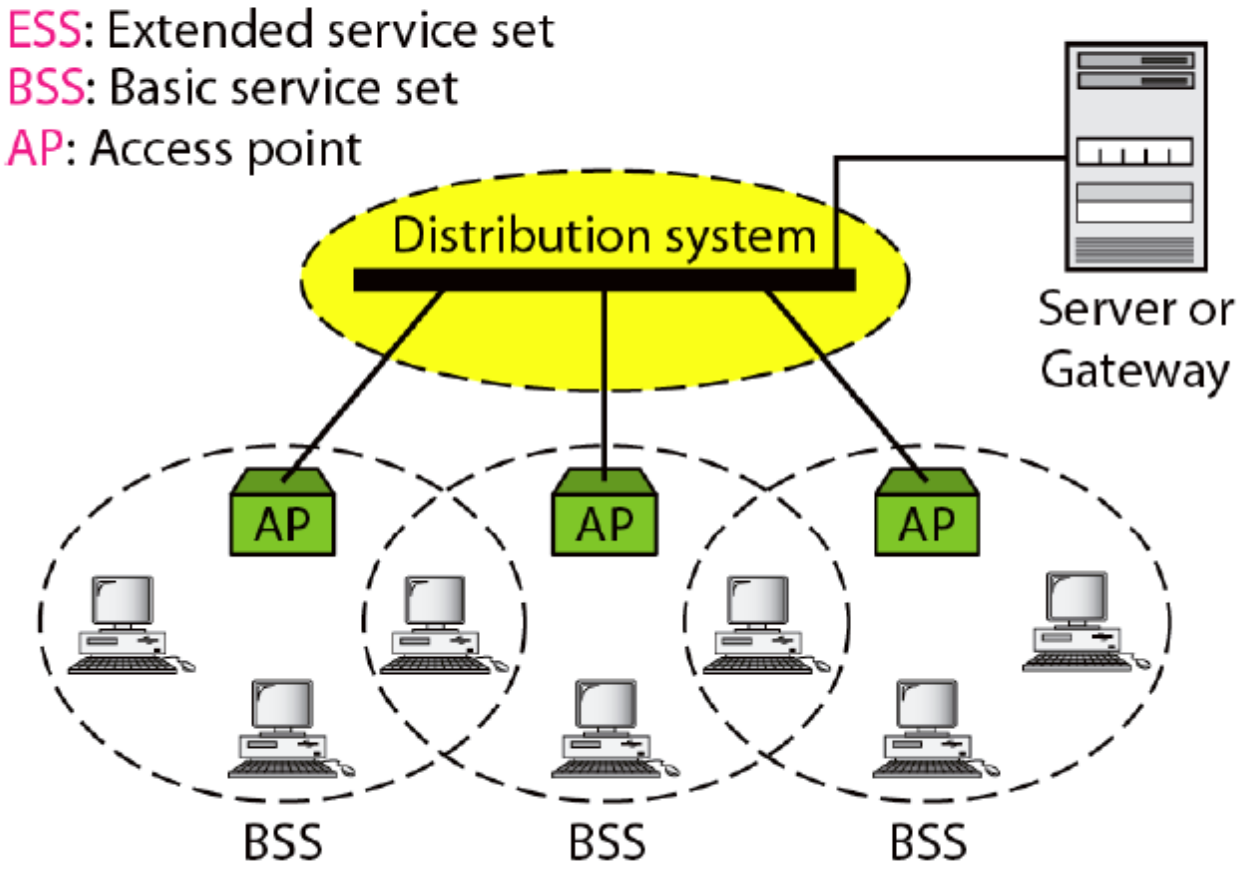
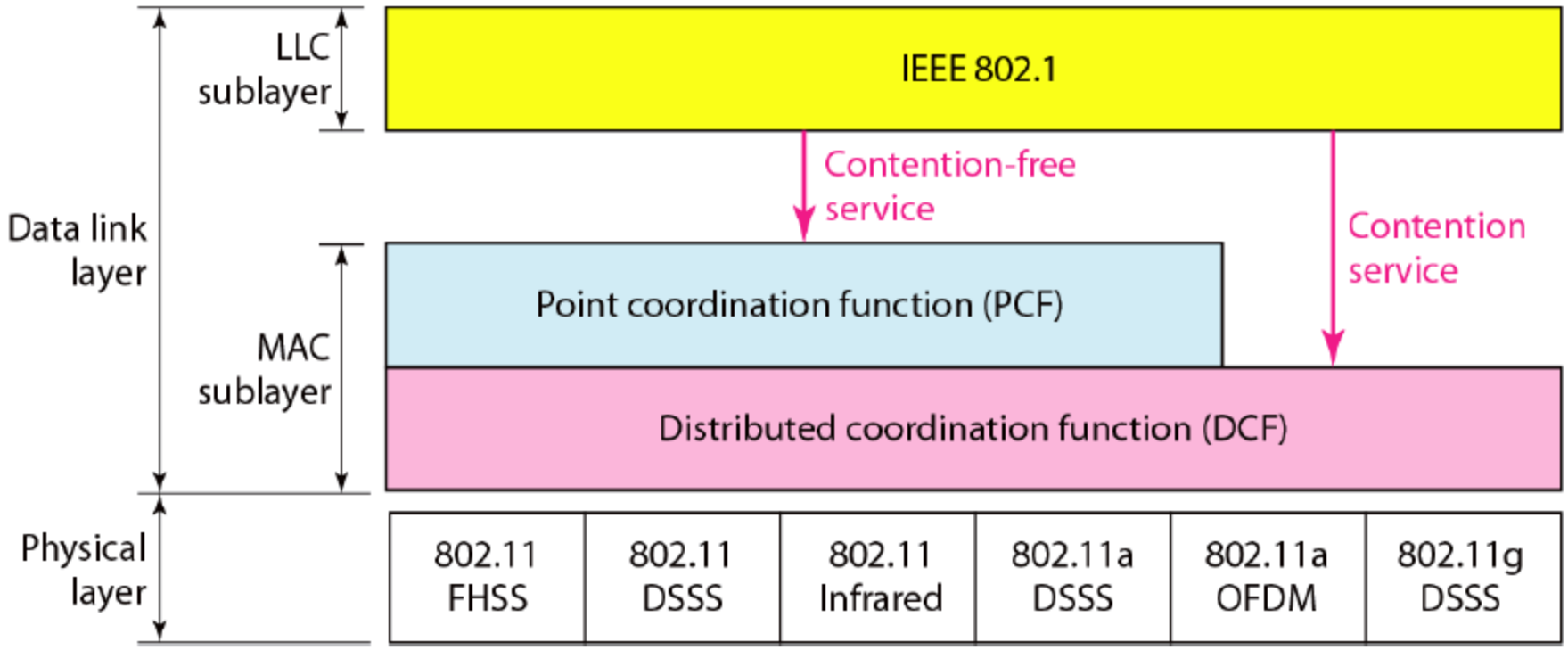


Figure 14.3 *MAC layers in IEEE 802.11 standard*



802.11 Physical Layer

Table 14.4 *Physical layers*

IEEE	*Technique*	*Band*	*Modulation*	*Rate (Mbps)*
802.11	FHSS	2.4 GHz	FSK	1 and 2
	DSSS	2.4 GHz	PSK	1 and 2
		Infrared	PPM	1 and 2
802.11a	OFDM	5.725 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2.4 GHz	PSK	5.5 and 11
802.11g	OFDM	2.4 GHz	Different	22 and 54

802.11 MAC sub layer

802.11 supports two modes of operation:

1. Distributed Coordination function

2. Point Coordination function

Sending a frame using CSMA/CA

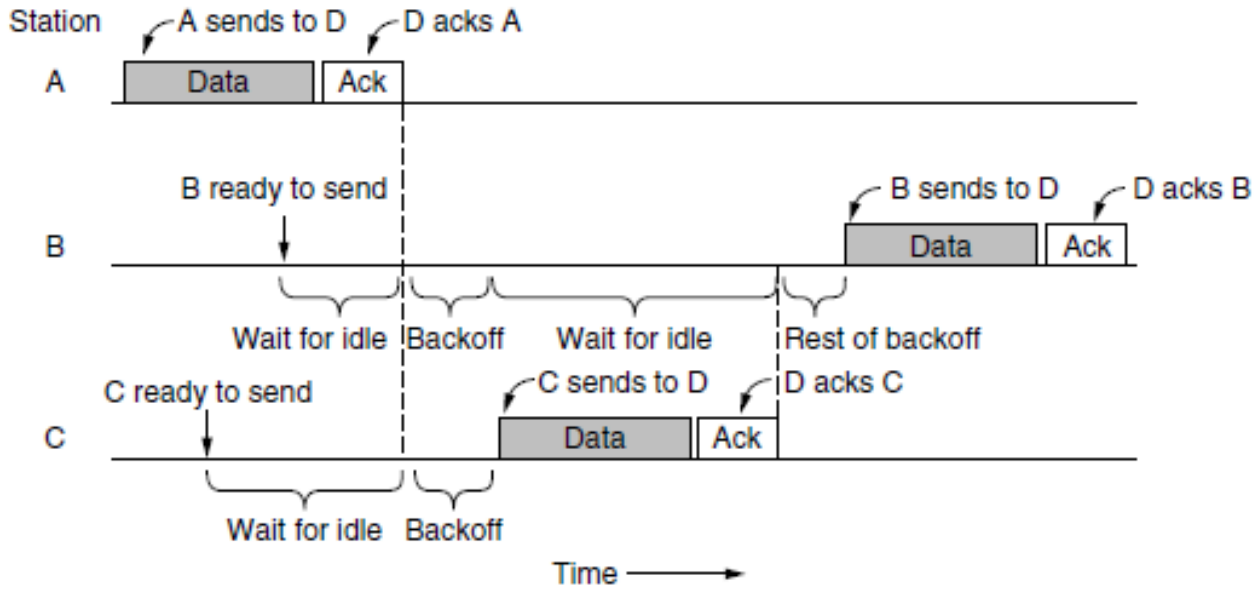


Figure 4-25. Sending a frame with CSMA/CA.

Virtual channel sensing using CSMA/CA

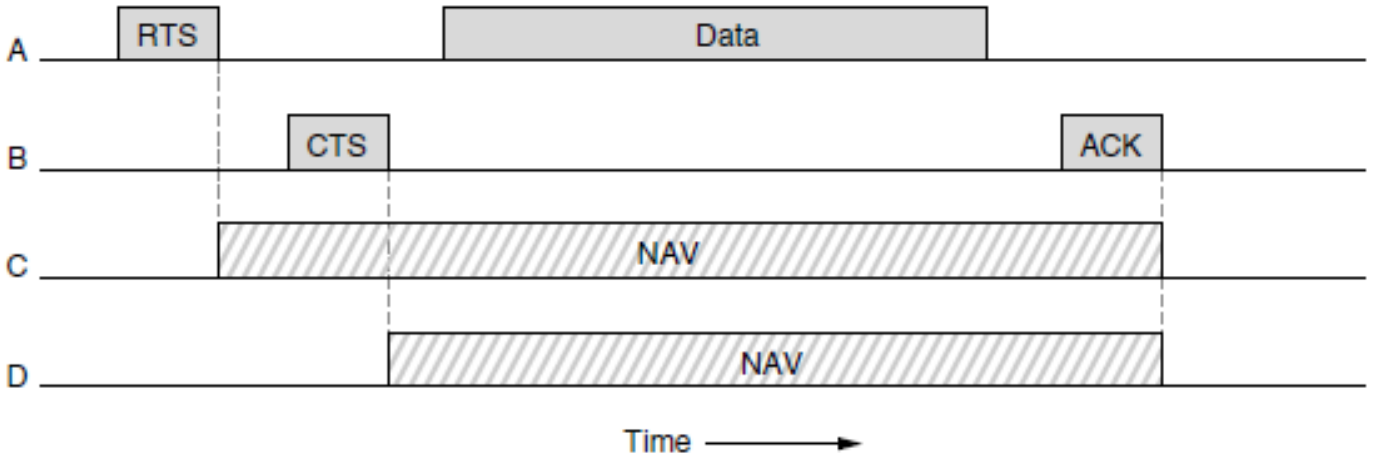
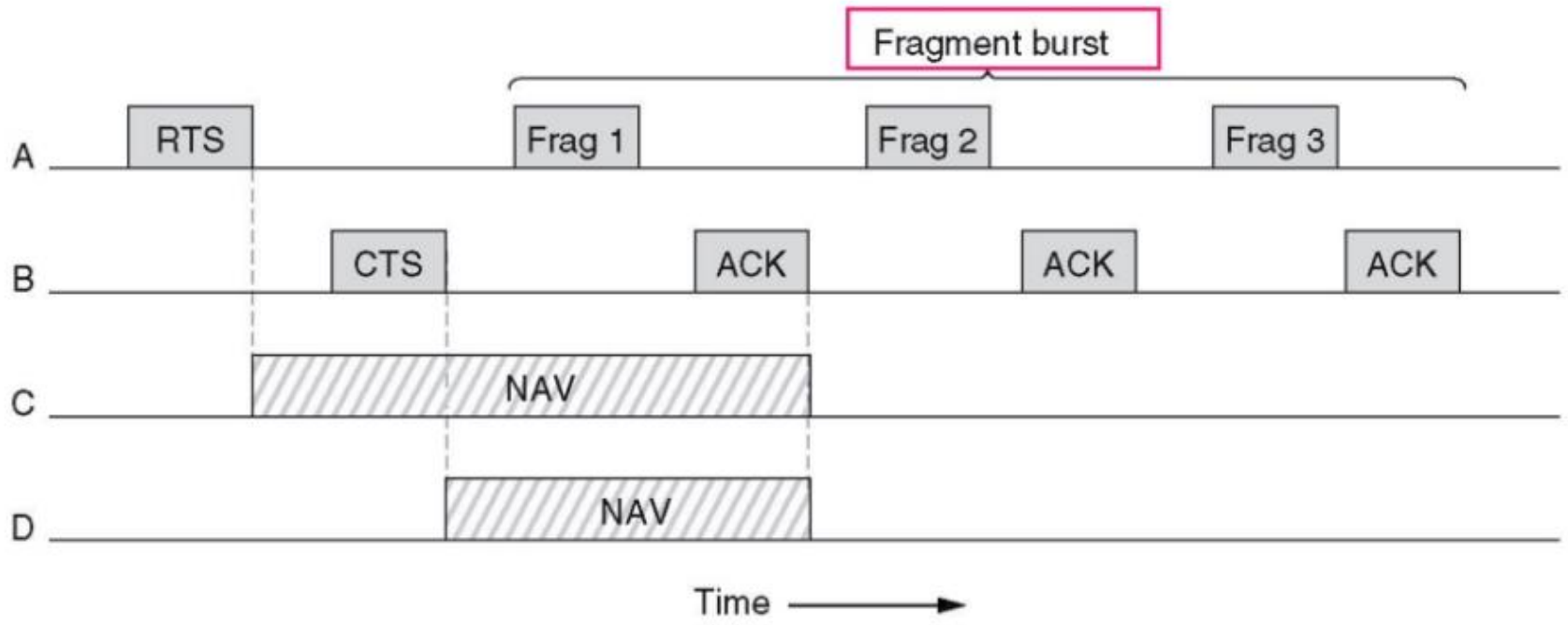


Figure 4-27. Virtual channel sensing using CSMA/CA.

Fragment burst

DCF-The use of virtual channel sensing using CSMA/CA.



stop-and-wait protocol

Interframe spacing in 802.11

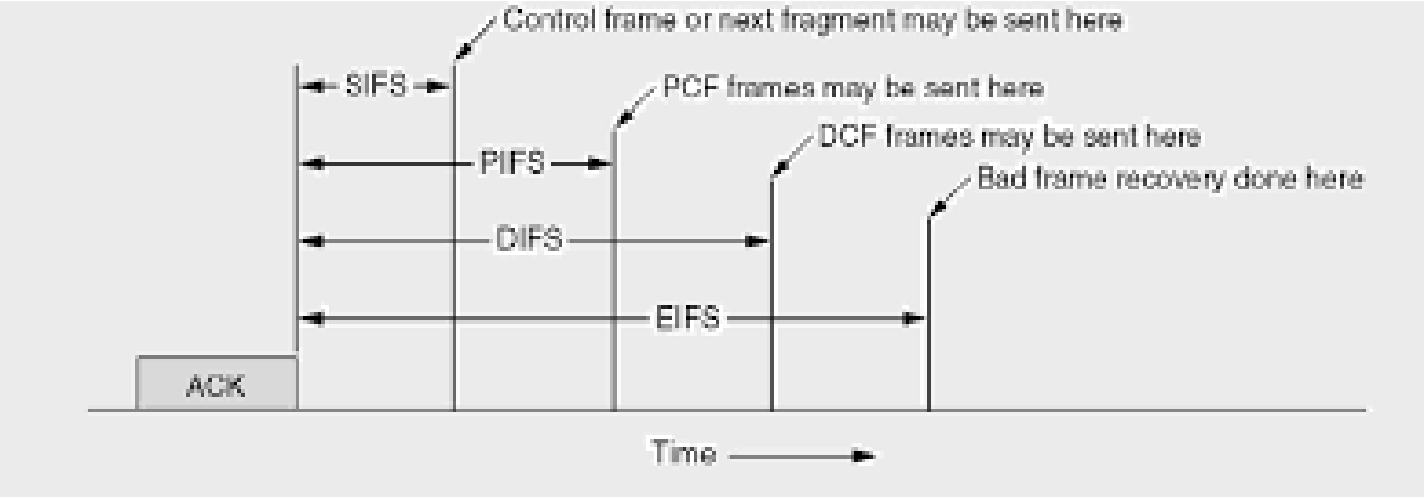


Figure 14.5 *CSMA/CA and NAV*

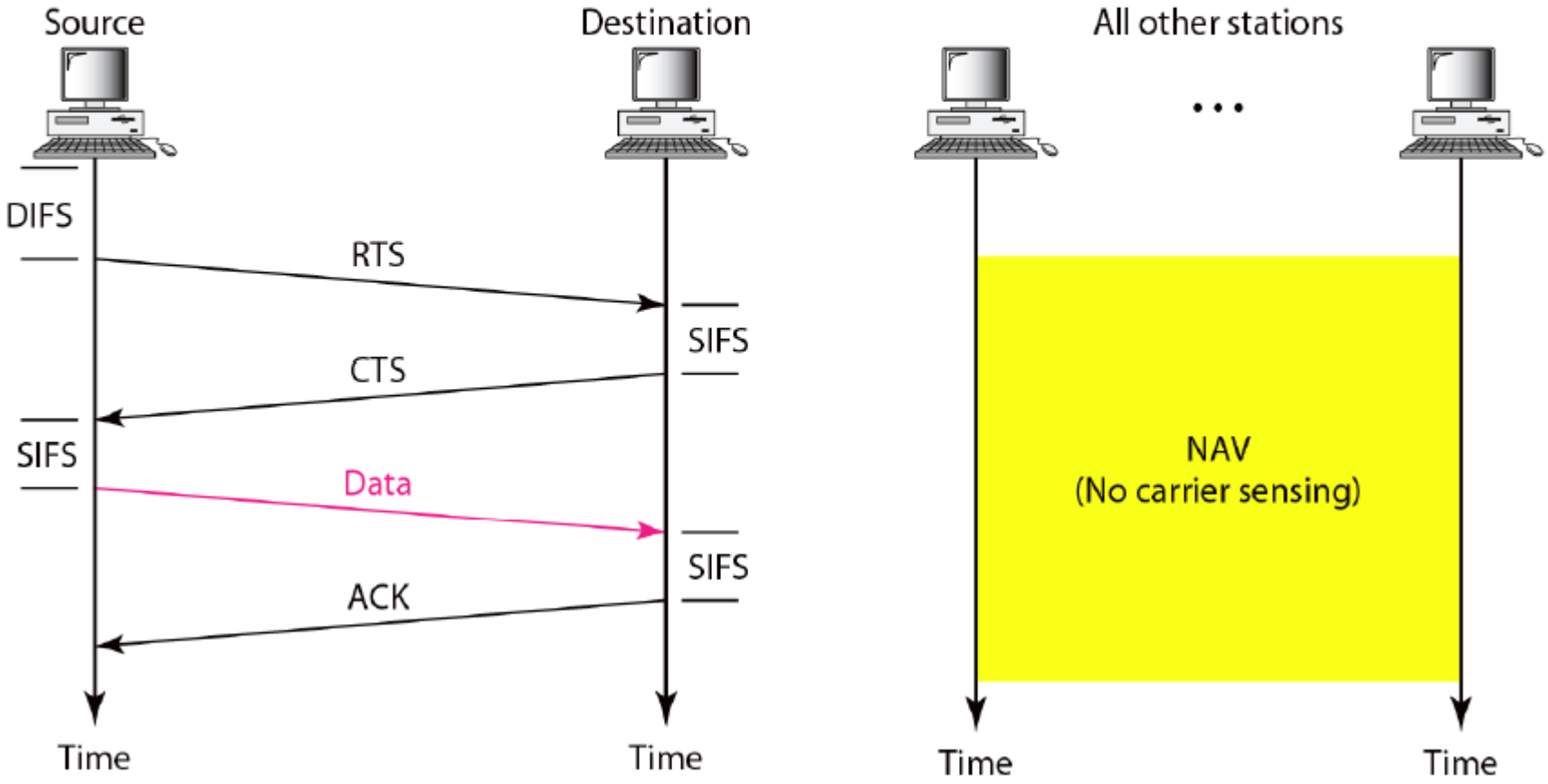


Figure 14.4 *CSMA/CA flowchart*

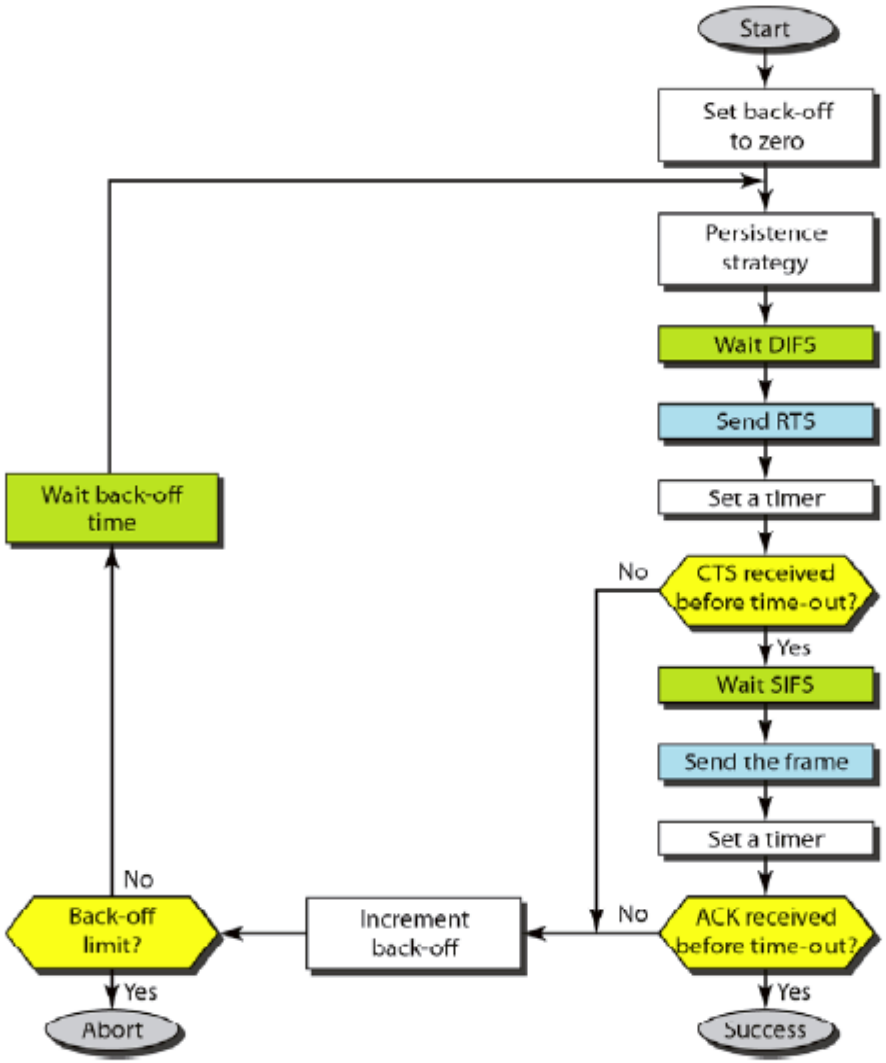


Figure 14.6 *Example of repetition interval*

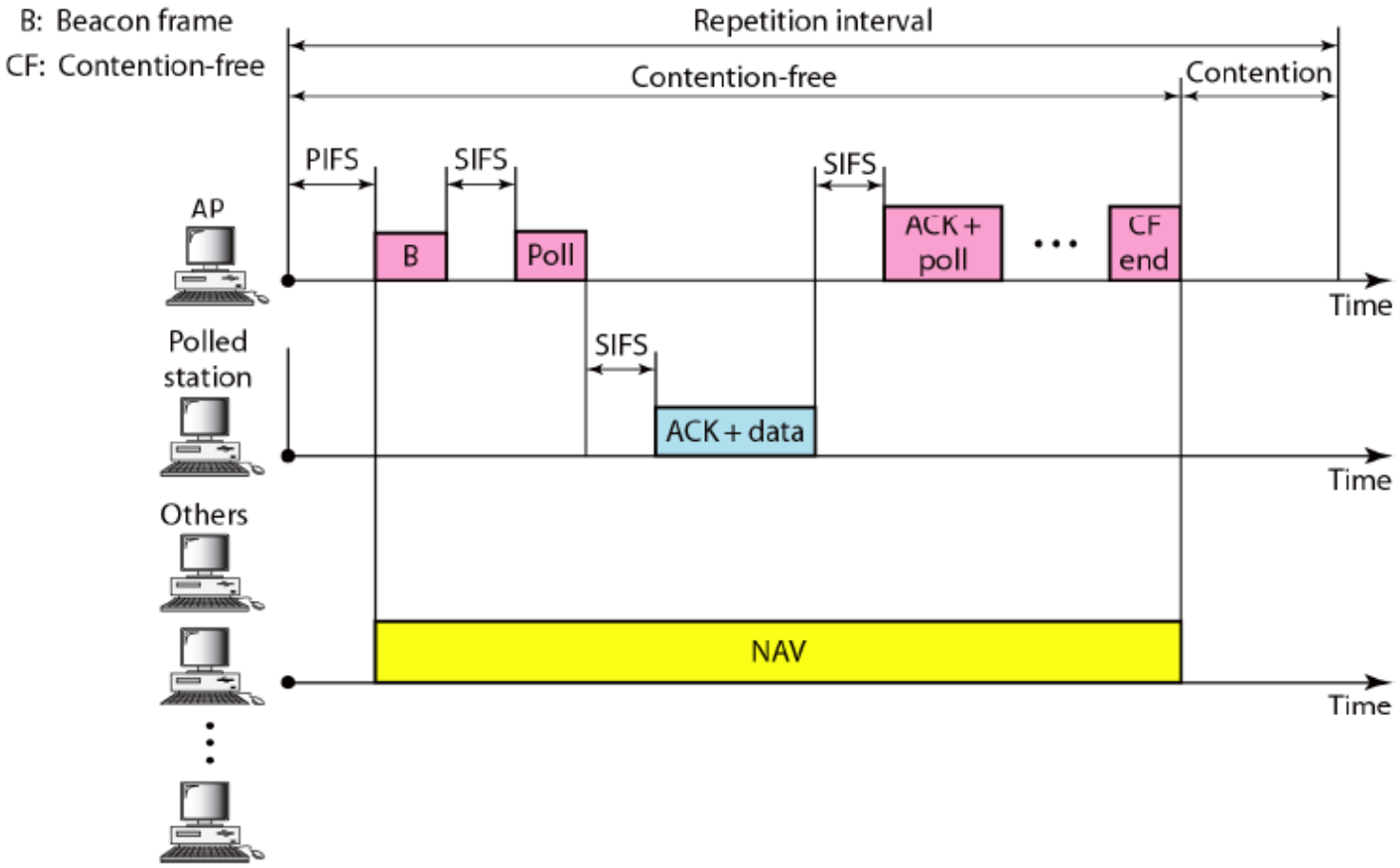


Figure 14.7 *Frame format*

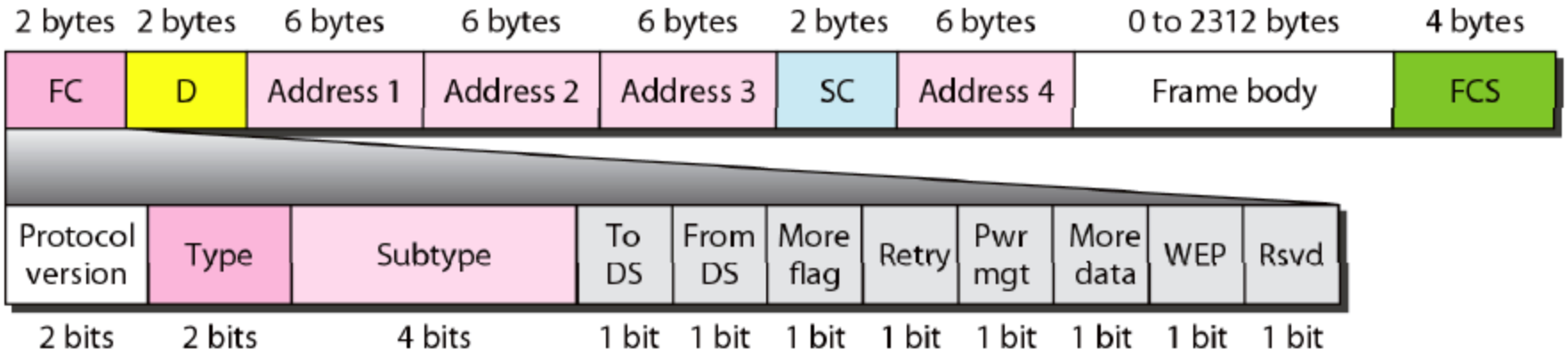


Table 14.1 *Subfields in FC field*

Field	*Explanation*
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 14.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

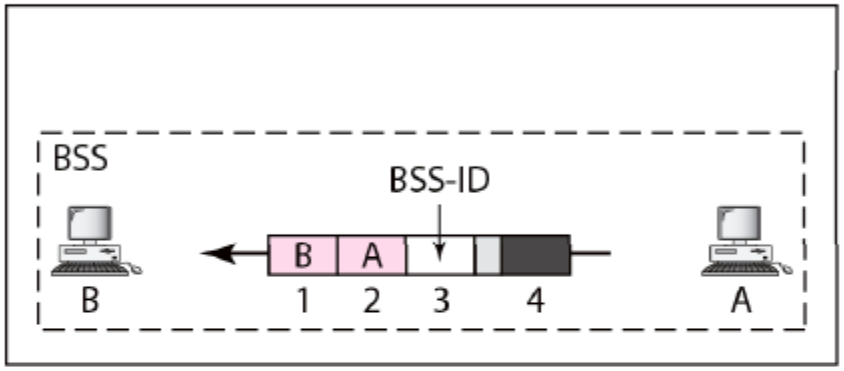
Table 14.2 *Values of subfields in control frames*

Subtype	*Meaning*
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

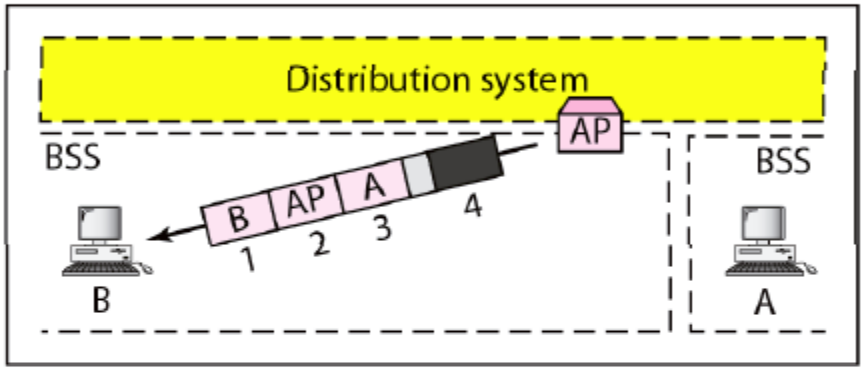
Table 14.3 *Addresses*

To DS	*From DS*	*Address 1*	*Address 2*	*Address 3*	*Address 4*
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source

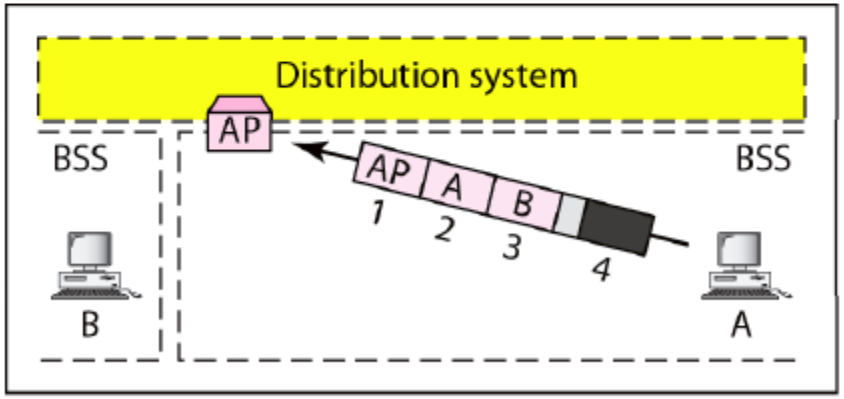
Figure 14.9 *Addressing mechanisms*



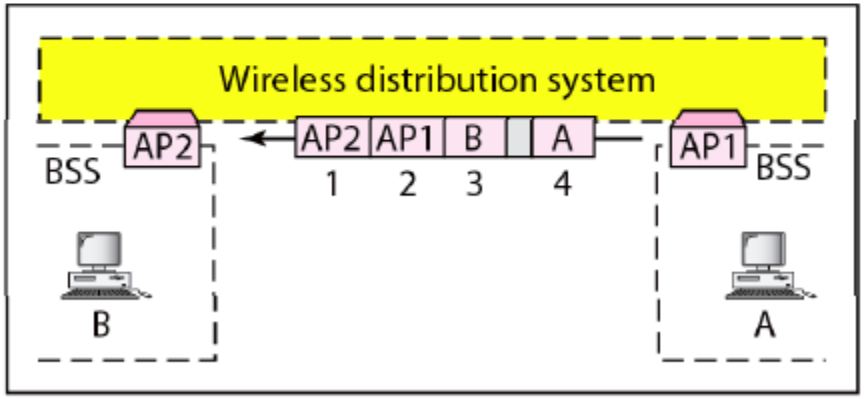
a. Case 1



b. Case 2



c. Case 3



d. Case 4

Bluetooth

Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers (desktop and laptop), cameras, printers, coffee makers, and so on.

A Bluetooth LAN is an adhoc network, which means that the network is formed spontaneously;

Figure 14.19 *Piconet*

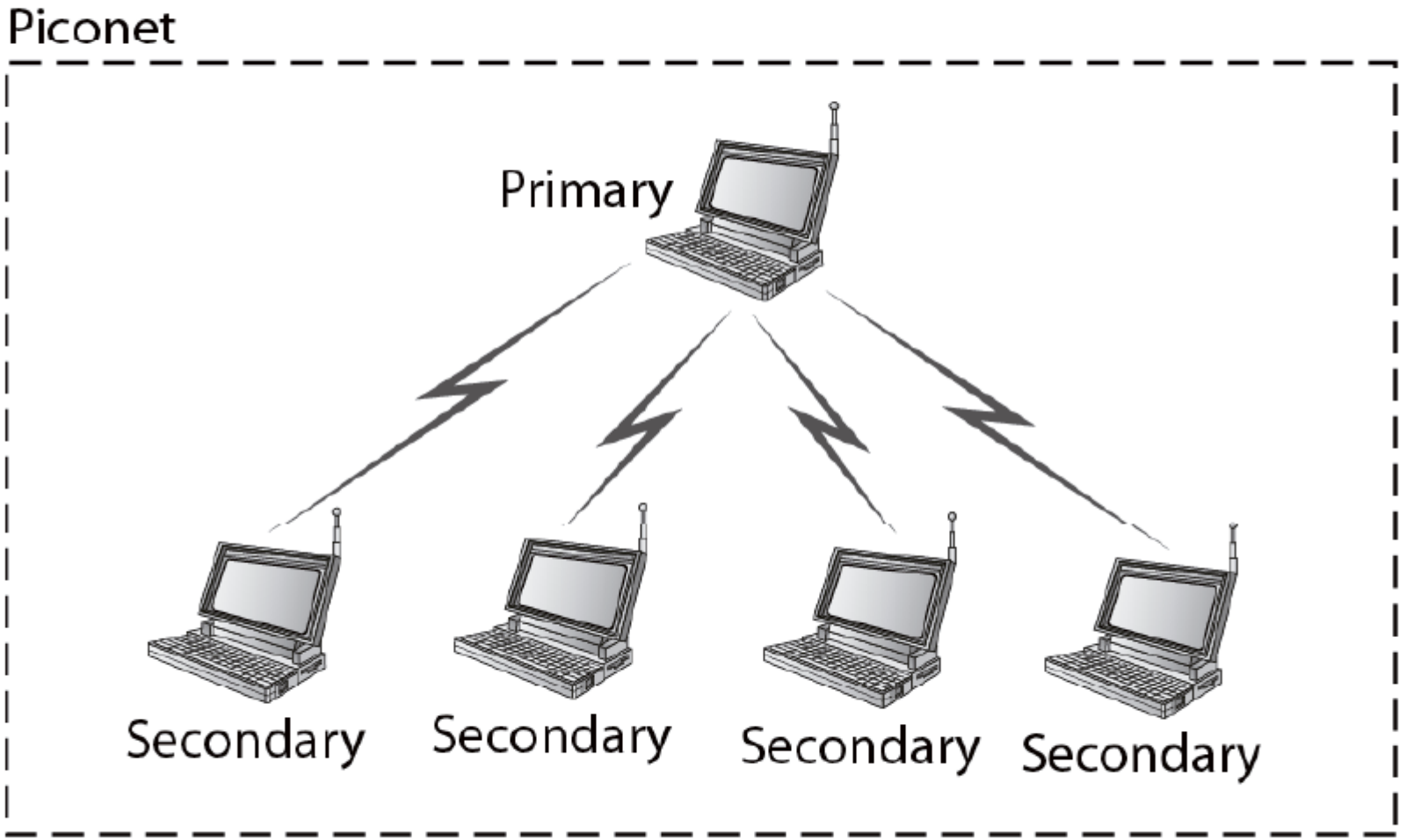
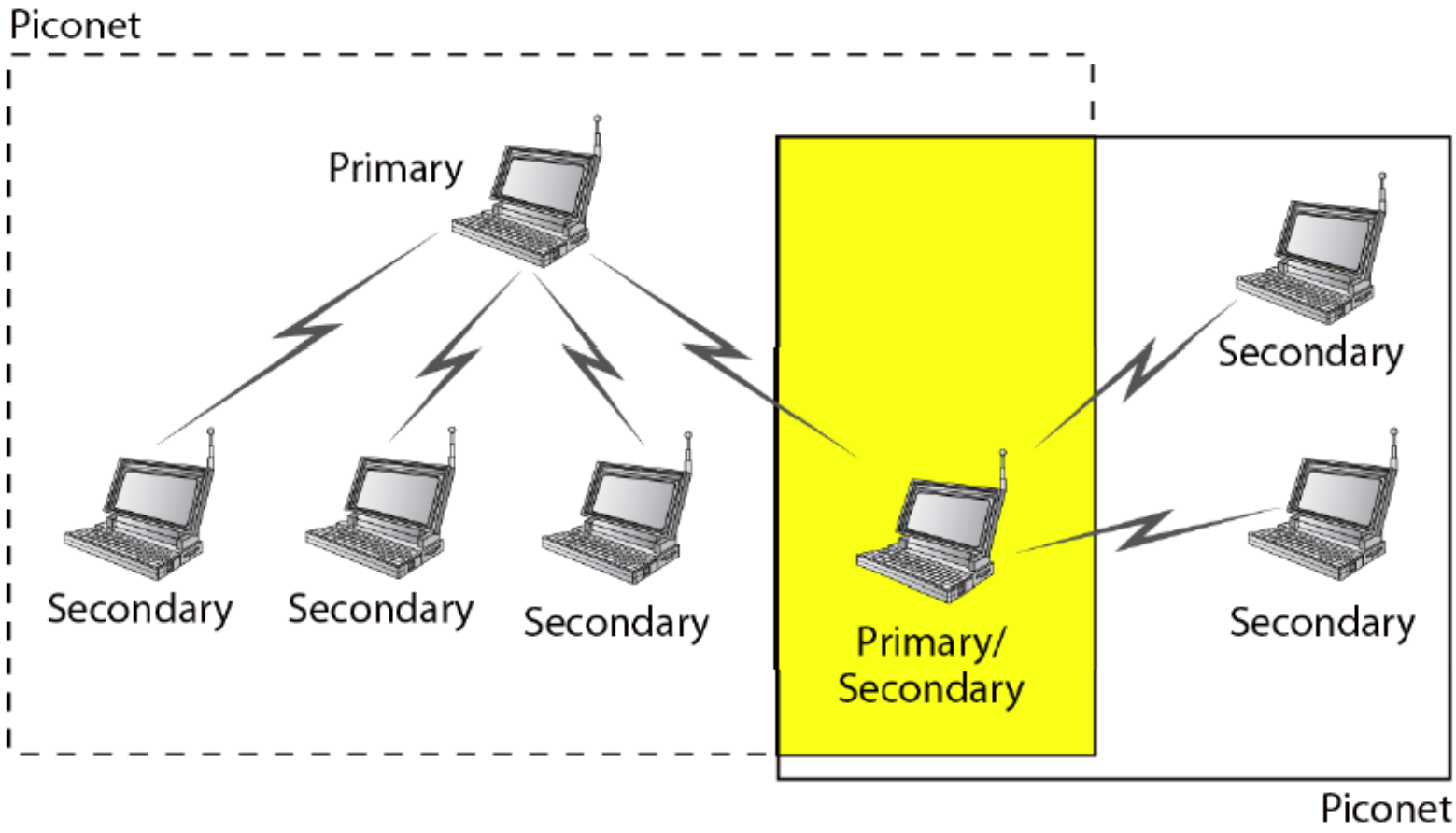


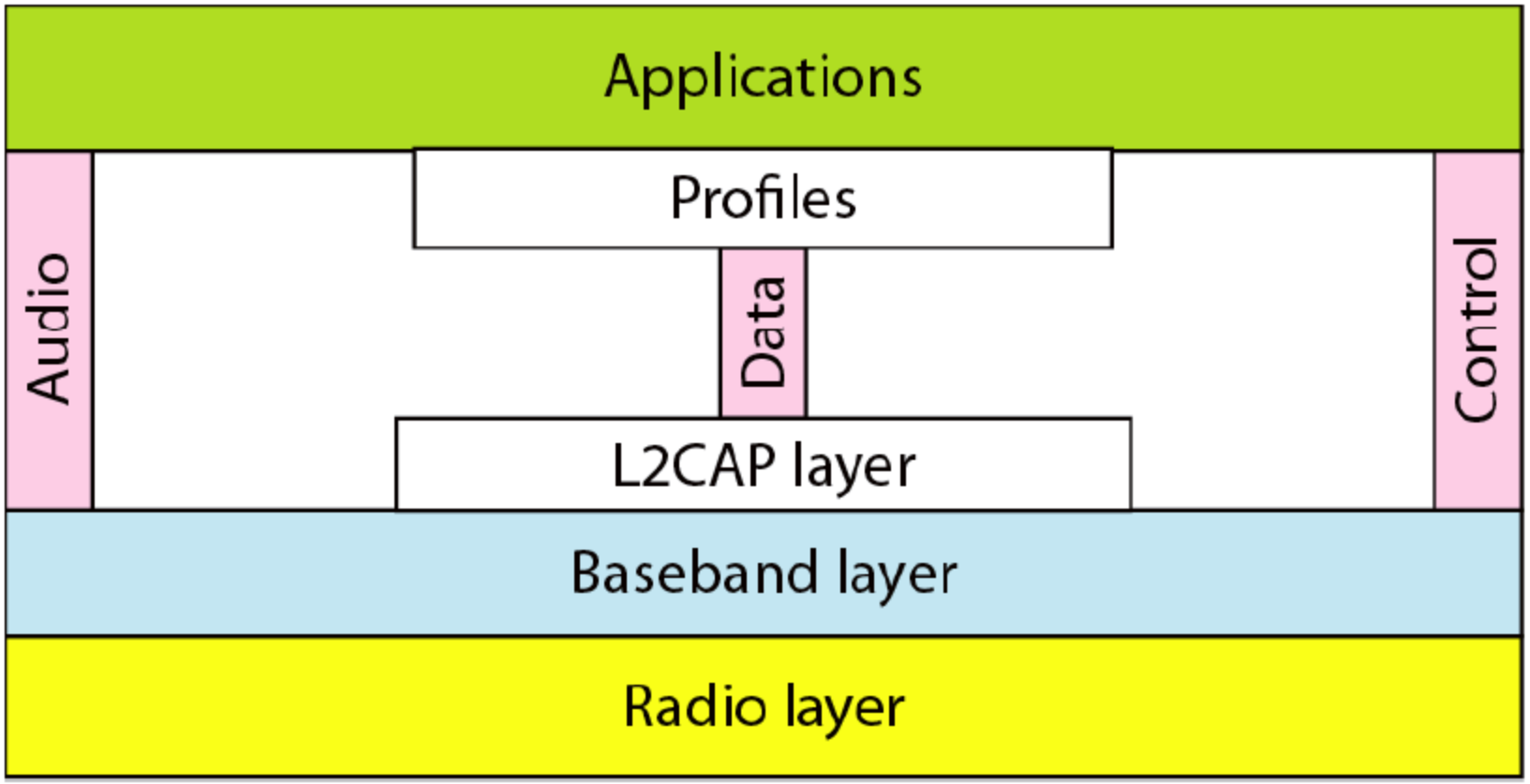
Figure 14.20 *Scatternet*



Who started Bluetooth?

- Bluetooth Special Interest Group (SIG)
- Founded in Spring 1998
- By Ericsson, Intel, IBM, Nokia, Toshiba;
- Now more than 2000 organizations joint the SIG

Figure 14.21 *Bluetooth layers*



Radio layer

The radio layer is roughly equivalent to the physical layer of internet model.

Bluetooth devices are low-power and have a range of 10m.

Bluetooth uses a 2.4-GHz ISM band divided into 79 channels of 1 MHz each.

Radio layer cont..

Bluetooth uses the frequency- hopping spread spectrum (FHSS) method in the physical layer to avoid interference from other devices or other networks.

Bluetooth hops 1600 times per second, which means that each device changes its frequency 1600 times per second.

A device uses a frequency for only 625μs(1/1600) before it hops to another frequency.

Baseband layer

The radio layer is roughly equivalent to the MAC sublayer LANs.

The access method is TDMA.

The primary and secondary communicate with each other using time slots. The length of a time slot is 625µs.

During this time slot either a primary sends a frame to secondary or secondary sends a frame to primary.

Figure 14.22 *Single-secondary communication*

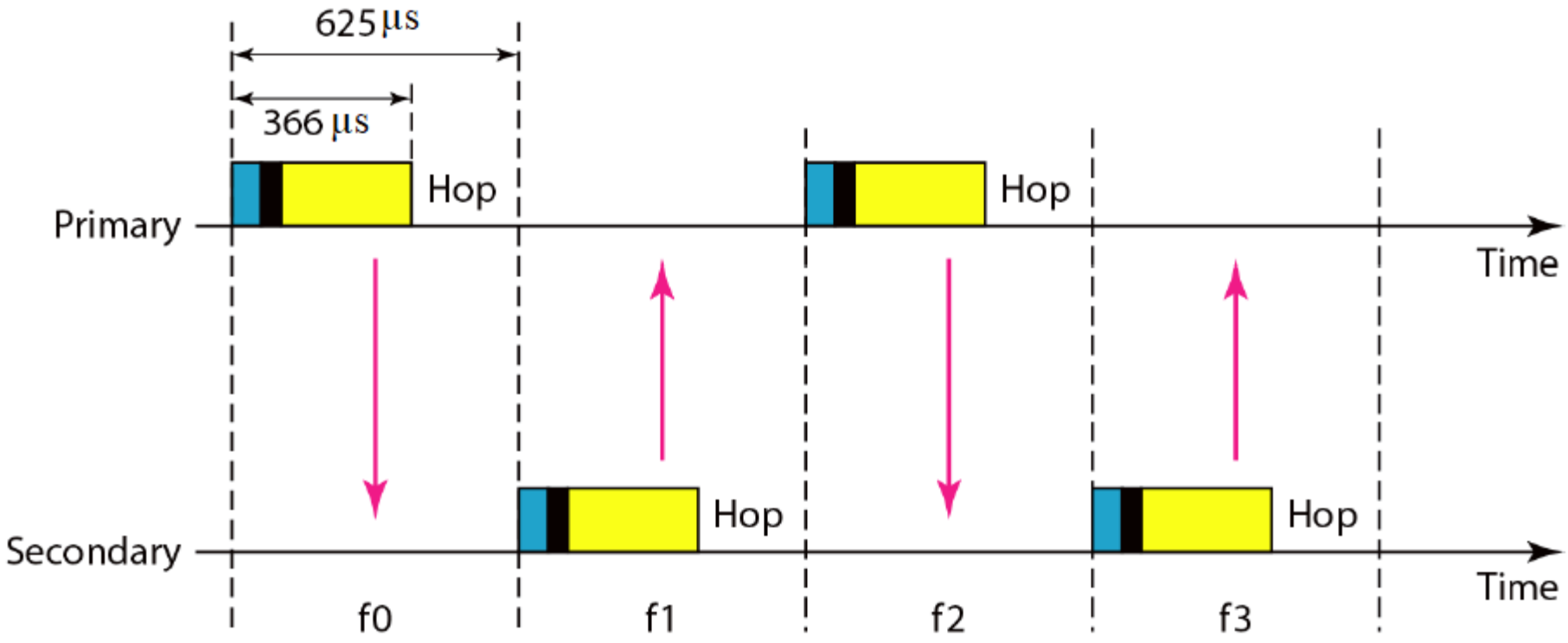
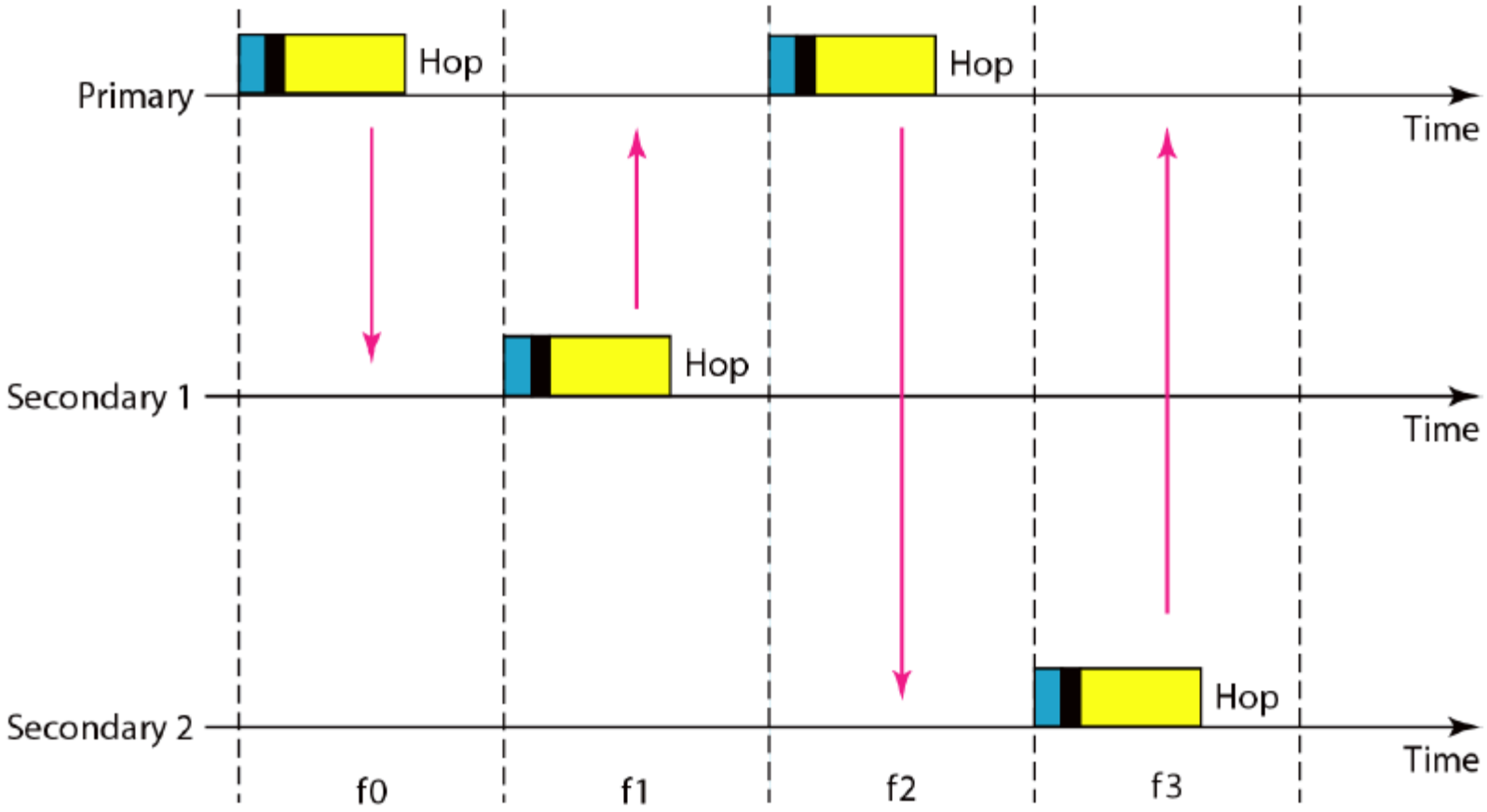


Figure 14.23 *Multiple-secondary communication*



Physical Links

Two types of links can be created between primary and secondary:

1. SCO: (Synchronous connection-oriented) link is used when avoiding latency (delay in data delivery) is important.

In an SCO link, a physical link is created between primary and secondary by reserving specific slots at regular intervals.

If a packet is damaged, it is never transmitted.

SCO is used for real-time audio where avoiding delay is all important.

Physical Links

2. ACL: An asynchronous connectionless link: ACL link is used when data integrity is more important than avoiding latency.

If a frame is corrupted, it is retransmitted.

ACL can use one, three or more slots and can achieve a maximum data rate of 721 kbps.

Frame Format

A frame in the baseband layer can be of– one-slot, three slot, or five-slot.
A slot is of 625µs.

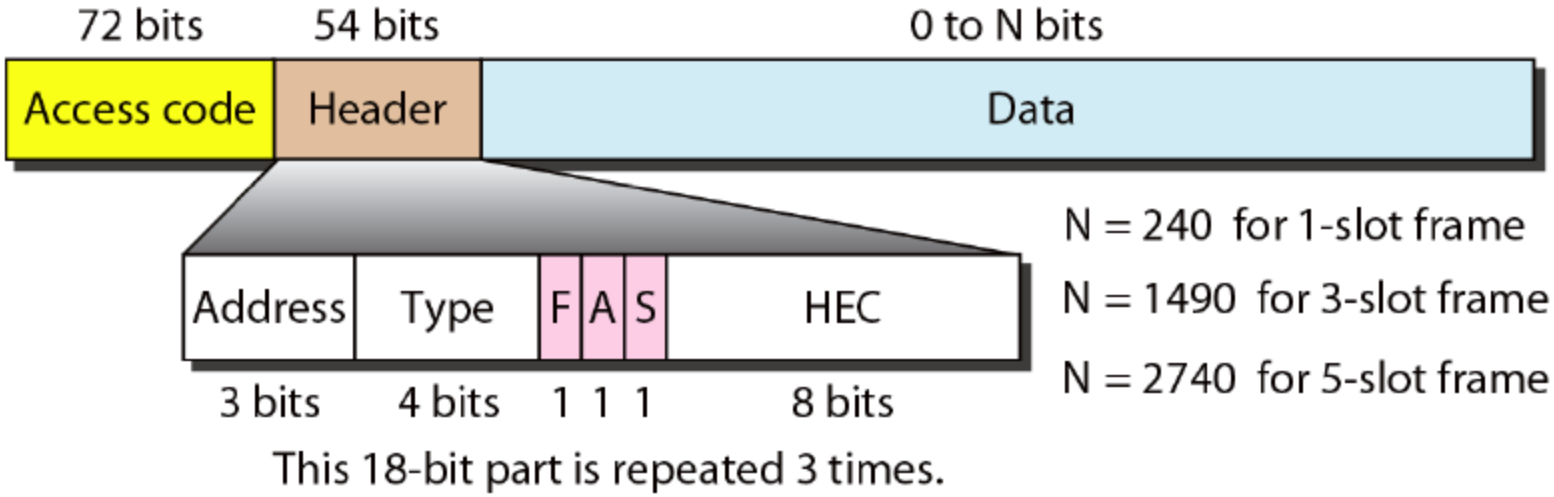
However in one frame exchange, 259µs is needed for hopping and control mechanisms.

This means that one slot frame can last only 625-259=366µs.
With a 1MHz bandwidth, the size of a one slot frame is 366bits.

For 3-slot: length of the frame is 3*625-259=1616µs or 1616 bits.

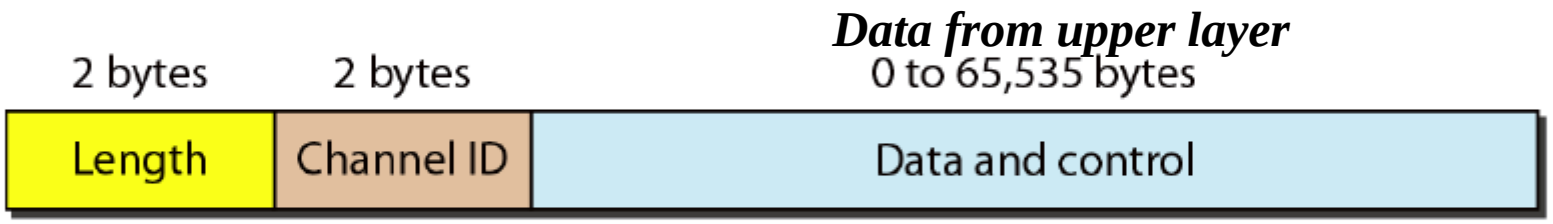
For 5-slot: length of the frame is 5*625-259=2866µs or 2866 bits.

Figure 14.24 *Frame format types*



N=366-72-54=240 (1 slot)
N=1616-72-54=1490 (3 slot)
N=2866-72-54=2740 (5 slot)

Figure 14.25 *L2CAP data packet format*



Duties of L2CAP layer

Multiplexing
Segmentation and Reassembly
QoS
Group Management